

CLINICAL ENDOCRINOLOGY & METABOLISM: COMPREHENSIVE GUIDE

CHAPTER 12: ACUTE AND CHRONIC COMPLICATIONS OF DIABETES MELLITUS

12.1 INTRODUCTION AND PATHOPHYSIOLOGY

Diabetes mellitus represents a group of metabolic disorders characterized by persistent hyperglycemia resulting from defects in insulin secretion, insulin action, or both.

12.2 ACUTE METABOLIC COMPLICATIONS

1. Diabetic Ketoacidosis (DKA): Characterized by absolute insulin deficiency, severe hyperglycemia (>250 mg/dL), and metabolic acidosis.
2. Hyperosmolar Hyperglycemic State (HHS): Predominantly in Type 2 Diabetes, characterized by extreme hyperglycemia (>600 mg/dL) and hyperosmolality (>320 mOsm/kg).

12.3 CHRONIC MICROVASCULAR COMPLICATIONS

1. Diabetic Nephropathy: Leading cause of end-stage renal disease (ESRD). Starts with persistent microalbuminuria.
2. Diabetic Retinopathy: Non-proliferative (microaneurysms, cotton-wool spots) and proliferative (neovascularization).
3. Diabetic Neuropathy: Distal symmetrical sensorimotor polyneuropathy (glove-and-stocking distribution), autonomic neuropathy, and mononeuropathies.

12.4 MACROVASCULAR COMPLICATIONS

Coronary artery disease, cerebrovascular disease (ischemic stroke), and peripheral arterial disease (PAD).